INTERNSHIP REPORT

A Governed Intelligence for Islamic Banking

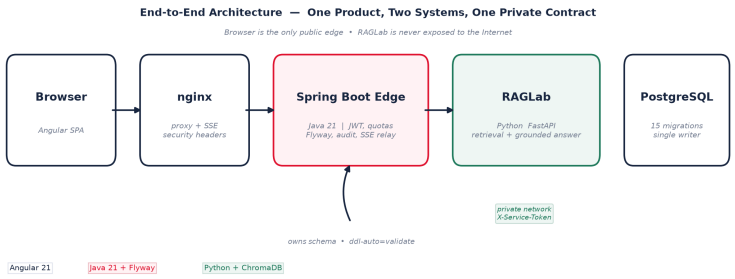
The Integrated RAGLab & Bank Assistant Platform

Every answer cites its source. Every refusal explains why.
A bank-operated web application consuming a governed RAG intelligence engine.

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Supervisor	Mrs. Ahlem Ben Haddoud - Advisor, Al Baraka Bank
Academic year	2025 - 2026

Internship report - No account identifiers, repository URLs or credentials are included by design.

All figures are traceable to frozen benchmark artifacts in the two project repositories.



Dedication

To my family, for their constant support,
to Mrs. Ahlem Ben Haddoud, for the guidance, rigor and trust that shaped this work,
and to everyone who believes technology, when governed, can serve trust.

Farah AYARI - ESPRIT, Software Engineering

Acknowledgments

My deepest thanks go to Mrs. Ahlem Ben Haddoud, internship supervisor at Al Baraka Bank Tunisia - Centre Urbain Nord, for her availability, clear decisions and thoughtful feedback at every key step.

I thank the teams of Al Baraka Bank for their welcome and for the documents and operational feedback that revealed defects invisible in tests. I thank the faculty of ESPRIT for the education that made this project possible.

Abstract

In regulated banking, a wrong answer about a rate, fee or deadline is a legal risk. This internship delivered a complete, governed answer: an integrated platform where a bank-operated web application consumes a dedicated RAG intelligence engine, both built to never invent figures or rules. The report covers the architecture, the major technical choices, the problems that were solved and the results that can be measured. No account identifiers or secrets appear in it by design.

Keywords: RAG - Islamic Banking - Grounded Answers - Citation Verification - Arabic NLP - Spring Boot - Angular

Conte

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Reading guide: Ch.1-3 set the context (who/why/how it fits). Ch.4-5 are the technical core (engine + app). Ch.6 is the most important - it explains what broke and how it was fixed. Ch.7-9 close with security, deployment and measured results.

CHAPTER 1 - 1

Host Organization

Al Baraka Bank Tunisia - Centre Urbain Nord, Tunis

Identity and History

Al Baraka Bank Tunisia is the first Islamic bank in Tunisia and the Maghreb. Created on 15 June 1983 as Beit Ettamouil Tounsi Saoudi (BEST Bank) by the Tunisian State and Sheikh Saleh Abdallah Kamel (founder of the Dallah Al Baraka Group), it was renamed Al Baraka Bank Tunisia in 2009 as part of the Group's identity unification. In 2013 it became a universal resident bank, serving individuals, professionals and enterprises with Sharia-compliant products. The Group, Al Baraka Group B.S.C. in Bahrain Bay, runs 600+ branches in 12 countries. Today the bank operates 44 branches across Tunisia, headquartered at Centre Urbain Nord - elected Best Islamic Bank in Tunisia (2018, 2019, 2021) and honored with five IFN Awards in 2024.

Product Universe

All products are framed by Tunisian Law No. 2016-48 of 11 July 2016 on banks and financial institutions (Articles 12-16 define Islamic operations) and by Central Bank Circular BCT 2019-08 - the same two documents that form the corpus of this project.

Family	Products
Trading & Investment	Murabaha (cost-plus), Salam (deferred delivery), Istisna'a, Mudaraba/Musharaka, Sukuk, participative deposits
Leasing & Financing	Ijara with acquisition option, Beit Al Baraka (home), Tas'heelat (personal), equipment & working-capital
International & Daily	Documentary credits, remittances, leasing, everyday banking

Product families the assistant must explain without error.

The Internship Site

Al Baraka Bank - Centre Urbain Nord

Avenue Cheikh Mohamed Fadhel Ben Achour, Centre Urbain Nord, 1082 Tunis - Tel: (+216) 71 183 203 - Head office: +216 71 18 65 00 - SWIFT: BEITTNTT
This district is the bank's operational heart, where product, compliance and IT decisions meet - and where the project was supervised by Mrs. Ahlem Ben Haddoud.

Project Vision

A Governed Banking Intelligence

The Business Problem

Banks produce vast regulatory text: laws, central-bank circulars, internal guides. Finding a reliable answer is costly; giving it to a raw language model is risky - it always answers, even when it does not know. Three tensions drove the project: the user's everyday question rarely matches the document's technical wording (sigha, mahal al-aqd, tamwil); the corpus is Arabic but questions arrive in French and English; and a false rate or deadline is a legal exposure.

North Star (5 Properties)

#	Property	In Practice
1	Grounded & Auditable	Every answer traceable to a source; nothing fabricated
2	Compliant	PII scrubbed on output; no learning from chats
3	Quota-Controlled & Secure	Public channel metered; engine off the Internet
4	Regulable by Staff	Admins edit taxonomy, facts and documents
5	Multilingual (RTL)	FR / AR / EN; Arabic RTL first-class

Five properties governed every technical choice, in priority order.

What Was Delivered

> **Two Systems, One Product**
>
> RAGLab answers; Bank Assistant serves customers and governs what RAGLab knows. Neither writes into the other's store. Staff sees every conversation from its first message; customers never see the source document reference. That boundary makes the answer path auditable.

The web app provides a public landing page, a floating assistant widget, an authenticated client portal (accounts, cards, transactions, simulations, advisor appointments) and an 11-screen admin back-office with a 5-tab RAGLab Console. The intelligence engine is a stateless, versioned HTTP microservice with 25+ routes, a frozen contract and a full audit trail.

CHAPTER 3 - 7

System Architecture

The Full Picture - Browser to Intelligence

End-to-End Flow



End-to-end architecture: browser is the only public edge. RAGLab is never exposed to the Internet; it is addressed by ip:port on a private network.

In words: the browser talks only to nginx (Angular SPA, security headers, SSE). Nginx proxies /api to the Spring Boot edge (Java 21) which owns authentication, roles, quotas, persistence and admin. The edge calls RAGLab over an authenticated internal contract (X-Service-Token) and streams the answer back. PostgreSQL (Flyway, 15 migrations, single writer, ddl-auto=validate) holds the business schema. The vector index lives inside the Python agent - no shared-DB write coupling. The edge opens the exchange row before calling the engine, so even a failure is visible.

Why This Separation Is the Design

The split is the guarantee: the business schema can evolve without re-indexing; the index can be rebuilt without migrating the business database. An admin switching the engine, a document re-published or a model key rotated all flow through explicit, audited actions.

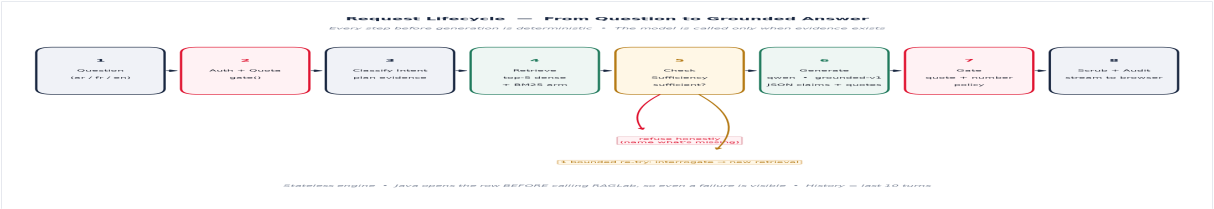
Technology Stack

Layer	Choice	Why
Frontend	Angular 21 (standalone), Vitest	SPA with lazy modules; no Google-fonts dependency
Backend Edge	Java 21, Spring Boot 4, Flyway, JWT, virtual threads	Owns schema with validate; avoids carrier pinning
Intelligence	Python 3.11, FastAPI + uvicorn	No LangChain - every behavior readable
Vector Store	ChromaDB local (cosine) + BM25	Local avoids cloud cost & banking-data egress
Embeddings / LLM	NVIDIA 2048-d + Qwen free	Measured best on Arabic; no fallback
Persistence	PostgreSQL, Redis queue, Flyway	Single writer; durable BRPOP queue
Deployment	Docker Compose, offline tarball	Same images dev -> prod; Windows path

Stack: sobriety as a governance decision.

Two Lifecycles

- Document: POST /documents (content-hash versioned; identical bytes = no-op; SHA-256 dedup) then batch POST /ingest (one job at a time; 409 is a state). The edge's documents table is a mirror - RAGLab owns the index and reports pending/indexed/stale honestly.
- Conversation: browser -> edge (quota gate) -> RAGLab (retrieval -> generation -> gate) -> edge (scrub -> persist -> settle -> stream).



Request lifecycle: from question to grounded answer. Every step before generation is deterministic; the model is called only when evidence exists.

CHAPTER 4 - 8

The Intelligence Engine - RAGLab

How It Thinks: From Scan to Grounded Answer

Corpus and the Arabic Challenge

Document	Role	Fragments
Law 2016-48 (198 articles)	Superior legal reference	233
BCT Circular 2019-08	Operational framing	13
Internal Islamic Operations Guide	Internal policy	54
Intro to Islamic Finance (Madkhal)	Training, definitions	39
Total	4 real Arabic documents	339

Governed corpus: 4 real documents -> 339 structure-aware fragments.

The PDFs are scanned legal Arabic: visual-order lines, reversed word order, corrupted digits, broken headings. Indexed directly, the corpus is noise. The fix is linguistic - see Chapter 6.

Structure-Aware Chunking

Two chunkers are measured. Legacy size mode (220 tokens + 40 overlap) remains; adopted restructure mode does three steps: (1) semantic normalization - gazette cleanup, repeated-header dropping (>=3), section-marker extraction (al-fasl/al-bab), visual-order repair by zone into Markdown; (2) breadcrumb enrichment - injecting h1 > h2 > h3 ancestry above every heading/table; (3) recursive structural split over ['\n# ', '\n## ', '\n### ', '\n\n', '\n', ' '] at 220/40. Each chunk carries an identity card: document, heading, language, section_type, chunk_fp (fingerprint v4), embedding_fp, model.

Impact

+20 points hit@1 over window splitting (live: 51 -> 71; offline BM25: 42 -> 62). This single change outranked every retrieval experiment.

Embeddings and Storage

- Model: nvidia/nemotron-3-embed-1b, native 2048-d, NVIDIA Build endpoint (batch 32, backoff + Retry-After, bisect on degenerate vectors, resumable cache).
- Store: one ChromaDB collection per (embedding-space x chunking mode), hnsw:space=cosine. BM25 lexical index in parallel for hybrid modes.
- Fingerprint guard: chunk_fp + embedding_fp stored per chunk; stale collection refused before any model call. No mixed indexes served.

Retrieval: Dense Wins

Mode	hit@1	hit@3	hit@5
vector (dense, default)	71	80	87
rrf (k=60)	71	80	82
blend (lambda=0.7)	62	80	84

Measured identically (live, 50 questions). Dense stays default; French gap is sharper: 50/67/67 vs 33/33/50.

Understanding the Question

Module	Function
intent.py	Rule-based intent; splits compound requests; detects ambiguity
evidence_plan.py	Derives evidence requirements - never the answer
sufficiency.py	Checks plan vs fragments: sufficient/insufficient/contradictory
interrogate + topic_map	Only when nothing covered: one bounded LLM call re-expresses the question (262-node map)

Delivery: Citation Gate and Audit

- Prompt grounded-v1: question + sources as untrusted data; no tools; strict JSON claims+evidence; quotes >=12 chars, verbatim.
- Three gates: (i) quote membership - normalized substring; (ii) numeric - every number in claim must appear in quotes; (iii) policy - source retrieved and allowed.
- Sufficiency commitment: nothing covered -> refuse before generation naming what's missing; partial -> one bounded regeneration. PII scrub after gate; audit JSONL via GET /audit (500 cap) correlated by X-Request-Id.

Knowledge Layer (Law 2016-48)

- 198 units (=198 articles, stable ID loi-2016-48:artNNN), 77 cross-refs + 1 anchoring relation, 53 number records (rates/deadlines/caps/sanctions) via GET /numbers, 6 governance axes.

Result

A unit-description index answers law questions at 100/100/100 vs 67/92/92 baseline - every law question, including cross-lingual, finds its article at rank 1.

CHAPTER 5 - 9

The Digital Bank - Web Application

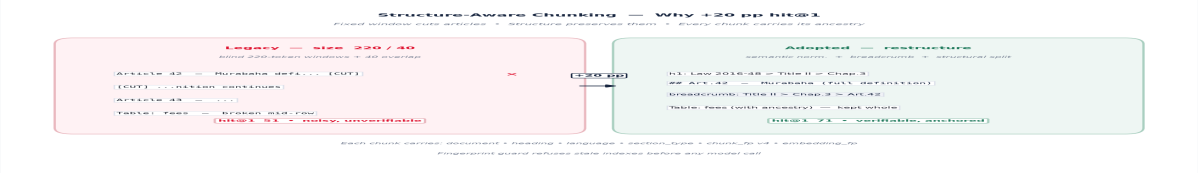
How It Serves: Portal, Auth and Control Room

Overall Shape

The web app is the only public edge. RAGLab is never on the Internet; it is addressed by ip:port on a private network with X-Service-Token. Frontend: Angular 21 standalone lazy modules (landing, chat widget, auth, portal, admin /admin/**). Backend: Spring Boot 4 Java 21 - JWT, Flyway, quotas, persistence, monitoring, 5-tab RAGLab Console. Database: PostgreSQL, Flyway single-owner V1..V15, ddl-auto=validate.

Customer Experience

- Landing at /: trilingual hero, products, assistant showcase, services, FAQ, footer - palette Red #E0122F / Orange #F5881C, RTL support. Floating widget with own session, quota-limited, no login needed.
- Auth split: Client vs Staff tabs, role-based redirect, /register. Portal /portal: accounts, cards, transactions, financing simulations, advisor appointments, profile + ChatGPT-style assistant with own history.



Structure-aware chunking: blind windows cut articles; restructure preserves them with breadcrumb ancestry (+20 pp hit@1).

Core Banking Data Model (V15)

Domain	Tables & Key Fields
Identity & Risk	clients (CIN, ABK######, KYC, risk, branch agence_id, employer, income)
Money	comptes (IBAN, RIB, solde), cartes (masked porteur 4571 **** **** 6842, no PAN/CVV), transactions
Products	financements (murabaha/ijara, margin, installments), beneficiaires
Service	advisor_requests (free-text), conversations, messages

Realistic core-banking shape. Transactions may contain third-party personal data (DPIA-relevant).

The assistant never sees raw balances - for an authenticated customer, a masked context (CustomerContextService) is derived at call time: slices and categories only, no identifiers leave the bank.

Authentication, Roles and Quotas

- JWT + server-side rotation: access <=1h; refresh with jti/family, single-use, reuse burns family -> 401. Revoked on logout or admin role change.
- Dynamic roles: admin_roles + permissions + audits (V14). 12 fixed permissions; SUPER_ADMIN reserved; unknown -> 403; every change audited with reason.
- Two-phase metering: gate() before engine, settle() after. Anonymous 3000/20, authenticated 10000/50, 5 sessions/hour. Per-session striped ReentrantLocks (64 stripes) avoid carrier pinning on JDK 21 virtual threads.

Streaming and Persistence

Streaming: Python stream -> Java invokeStream (pinned HTTP/1.1) -> Java SSE relay -> browser fetch+ReadableStream. Fallback to POST /message. Persistence: Java opens the row before calling RAGLab, persists after streaming, attributes engine=RAGLab and provenance, sends last 10 turns as context (turn_id + X-Turn-Id).

Admin Control Room

Screen	Function
/admin/facts	Governed facts CRUD - rates/fees outrank documents; append-only audit
/admin/taxonomy	Exemplar & release management; seal() + publish rollback
/admin/agent/health	Effective RAGLAB_URL, live probe latency
/admin/agent/quality	Answer -> retrieval -> quote existence checks
/admin/agent/documents	One drop zone, two chunk worlds (pending/indexed/stale)
/admin/agent/config	Models & keys (write-only, audited)
/admin/agent/journal	Every action, filterable, with motive
/admin/agent/monitor	Live feed + KPIs over last 200 exchanges

Every write audited, every read role-gated. Back-office stays in French (internal).

CHAPTER 6 - 10

Problems Solved

What Broke & How It Was Fixed - The Most Important Chapter

Not what was planned, but what broke - and what architecture prevents it from breaking again. Each fix is measurable and regression-tested.

Problem	Fix (Measurable)
Scanned Arabic PDFs in visual order, corrupted digits	restructure.py: NFKC + marker extraction + zone reconstruction; 29 batches (215 corrections), then 35/23/21; raw docs/ never modified; llm_repair_check.py gated
220-token window cuts articles mid-definition	Structure-aware chunking: +20 pp hit@1 (51->71 live; 42->62 offline) - largest win
Latin terms (murabaha) vs Arabic script	Multilingual embeddings + declared cross-script bridges (df-checked) + colloquial vocab + one bounded interrogation call
Guided retrieval dead in production (no search fn passed)	Live service now passes search callable; lesson: verify on the deployed service, not only in tests
Index duplication (859/1713 re-pushes)	RAGLab SHA-256 409 guard + content-hash versioning; batch POST /ingest as one job; in-house delete_source purge fix
French 'Quelle' inside 'Quelles' misclassification	Anchored regex + dedicated French procedural patterns; live-set regression tests
Streaming HTTP/2 Upgrade -> 422 on uvicorn	Pin java.net.http.HttpClient to HTTP/1.1 - one-line fix, days of debugging
Hardcoded token + 500 masking + nginx cache	Env wins over Compose default; network errors -> 502; index.html no-cache; all regression-tested

Eight hard parts - each with a measurable, audited fix. This table is the spine of the internship.

Two additional fixes worth noting: nginx cached the wrong index.html (now no-cache) and critical-CSS injected an onload handler blocked by CSP (now disabled). Both were invisible in unit tests and obvious in operation - the same lesson as above: reality is the only judge.

Security, Privacy & Governance

Defense in Depth for a Bank

Layers

Layer	Mechanism
Network	Only nginx public; backend+Postgres+RAGLab on private network; TLS + HSTS
Authentication	JWT (>=256-bit); X-Service-Token constant-time compare
Authorization	All admin @PreAuthorize; DB-backed 12 permissions; unknown -> 403
Input	Filename validation; 20 MiB cap -> 413; enum -> 400
Prompt Injection	Prompt marks Q+A as untrusted; local guard refuses live-data requests
Output	PII scrub after gate: EMAIL/PHONE/RIB-IBAN/CIN -> [LABEL]
Secrets	Env / platform secrets only; GET /keys shows presence; POST rejects placeholders

Each layer is small; together they are complete. No layer is trusted alone.

Personal Data

Nine families across clients/comptes/cartes/transactions/beneficiaires/financements/advisor_requests/conversations/refresh_tokens - from identifiers (CIN) to financial (IBAN) to free-text (chat, appointments). Two truths: (1) nothing identifiable leaves the bank beyond a masked slice/category context built at call time for an authenticated customer; (2) the highest real risk is free text - a question may contain an account number, stored verbatim and shown to staff. Output scrub + audit mitigate; retention/deletion lifecycle remains a documented pre-production gap for Legal.

No full PAN or CVV stored. Transactions may contain third-party personal data - DPIA-relevant. The Customer PII Report lists every field, who writes it and who may see it.

Audit & Threat Pass

- Every answer carries citations + engine attribution (RAGLAB_*); every admin action to agent_admin_audit with mandatory reason; every ingestion to JournalIngestion; every conversation visible from its first message; KPIs over last 200 exchanges.
- 20-point checklist (THREAT_PASS_6.2.md) drove fixes: second unauth door removed, constant-time compares, hardening headers, Swagger off in prod, host ports to 127.0.0.1. Remaining: magic-byte sniffing, per-request DB auth, Redis auth, nightly purge of expired refresh tokens.

Deployment & Operations

One Image Set, Three Environments

	dev	staging	prod
How	docker compose up	same images, real TLS	same images, HA stores
Profile	(default)	prod	prod
Swagger	on	off	off
Logs	plain / json	ecs	ecs
Secrets	.env (local)	platform secrets	platform secrets
Data	throwaway	anonymized monthly	live

Promotion moves digests dev -> staging -> prod; never rebuild for prod.

deploy/promote.sh enforces: refuses dirty tree, tags with commit SHA (never latest), verify compares GET /actuator/info + /version.json against HEAD. docker-compose.prod.yml sets prod profile and secrets. Kubernetes translation is 1:1.

Topology

- Internet --443(TLS,HSTS)--> Edge Proxy --80(private)--> nginx SPA --/api--> Spring Boot --> RAGLab (private) --> Postgres (PITR: daily full + WAL, 30-day retention, quarterly drill). Only the edge proxy is public.
- Flyway runs at boot; V1..V15 expand->migrate->contract; non-idempotent V1 was rewritten and baselined at 0.

Deploy Procedure

```
TAG=v1.0.0 docker compose -f docker-compose.yml -f docker-compose.prod.yml pull
docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d backend
docker compose logs backend | grep -E 'Flyway|Migrating|Successfully'
docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d frontend
E2E_BASE_URL=https://assistant.example.tn ./e2e/run.sh
```

Observability

- Correlation: turnId minted in Java, sent as body + X-Turn-Id, bound to MDC + Python contextvar - one grep spans both services; X-Request-Id echoed.
- Metrics: GET /actuator/prometheus (chat_turns_total) + GET /metrics (agent_turns_total). OpenTelemetry is the next step.
- Runbooks: RUNBOOK.md + HOST_TESTING.md + docker-compose.test.yml give a one-command workstation test (seeded staff + 4 customers with accounts/cards/movements/financings).

Results & Limitations

Measured, Not Claimed

How Quality Is Measured

- questions_50.json - 50 cases: 10 verbatim, 20 paraphrases, 15 cross-lingual (6 FR + 9 EN), 5 out-of-scope. 32 AR / 8 FR / 10 EN. Expected substring verified as distinctive.
- questions_targets.json - 10 AR cases: 2 per category (colloquial, synonyms, implicit, compound, ambiguity). Rules: paired in same run; +-2 pp across runs; frozen set never edited to improve a result.

Headline Results

What	Result
Chunking (live, 50)	size 51/64/71 -> restructure 71/80/87 (hit@1/3/5) - +20 pp
Retrieval (restructure)	dense 71/80/87 > RRF 71/80/82 > blend 62/80/84
Reranker (offline)	62/82/87 -> 62/89/89 - activated
Knowledge index (law)	67/92/92 -> 100/100/100 - every article at rank 1
Target categories	compound 100 | implicit+ambiguous 50/100 | synonyms 50 | colloquial 0
Model independence	agreement 0.720; Jaccard 1.000 (divergence is accept/refuse)
Final campaign (7+12+6)	retrieval 5/7 | topic map 5/5 | 3 answered / 3 refused, zero errors
Robustness (6)	Fabricated/borrowed/computed/unknown/injection -> all rejected
Tests & CI	472 unit + 133 offline | 8 workflows | 469 harness families | green

All numbers from frozen artifacts, not impressions.

Limitations

Limitation	Scope
Citation != entailment	Gate proves words/numbers come from source, not that claim logically follows
No threshold	Top-5 transmitted even if off-topic; safety = sufficiency + refusal
Single worker	One profile/index/ingest at a time; scale is designed, not deployed
Updates by re-ingestion	Old fragments serve until next ingest (now surfaced as stale)
Evaluation scale	50 + 10 cases small - not a security/legal guarantee

Readiness (Stated in the Repository)

Suitable for a supervised pilot on approved non-confidential documents; not ready for unsupervised customer-facing banking advice. This honesty is a product property - the system refuses rather than guesses, and its limits are documented where any auditor can read them.

Conclusion & Perspectives

What Was Built and What Remains

What Was Built

A working, documented and measurable platform: a complete language pipeline (reading and repair, structure-aware chunking, vector + lexical indexing, dense retrieval, intent and sufficiency reasoning, citation-gated generation, output scrubbing, audit) wrapped in a bank-operated web app that owns identity, quotas and persistence - all measured, versioned and traceable.

Lessons That Outlive the Code

- Measurement replaces opinion. Restructure over window, dense over fusion, lexicon off, expansion abandoned - all settled by paired numbers; several reasonable intuitions were rejected by data.
- Governance of decisions is architecture. Step-and-plan, activation gates and frozen sets let the project survive interruptions without silent regression.
- Reality is the only judge. The most expensive defects were invisible in tests and obvious in operation. Any claim about the deployed service must be checked on the deployed service.

Next Steps

- Per-document authorization and chunk-level topical metadata (path/level/parent_id) before any production re-index.
- Topical & faceted retrieval arms measured on frozen sets, without reintroducing regressions.
- Closing measurement campaign - re-measure every arm, document limits, then decide defaults.
- Industrialization: incremental updates, observability, adversarial evaluation, magic-byte validation, OpenTelemetry and a legal-reviewed retention/deletion lifecycle.
- Product: safely editable answer templates and conversation quality flags.

> *"The best system we built is not the one that answers the most questions,*
> *but the one that knows when not to answer - and can prove why."*
>
> Farah AYARI, ESPRIT - Al Baraka Bank, Centre Urbain Nord - Mrs. Ahlem Ben Haddoud

Refere

RAGLab/README.md	Supported pipeline, A/B tables
RAGLab/RAGLAB_SPEC.md	Reverse-engineered spec
RAGLab/RAGLAB_GAP_ANALYSIS.md	Gap analysis
RAGLab/raglab/CONTRACT.md	Frozen HTTP contract
bank-assistant/README.md	Web-app & governance
bank-assistant/PLAN_TO_PROD.md	Phased plan 0->10
bank-assistant/docs/DEPLOYMENT.md	Topology & promotion
bank-assistant/docs/CUSTOMER_PII_REPORT.md	PII inventory & gaps
bank-assistant/docs/THREAT_PASS_6.2.md	20-point checklist
RAGLab/docs/ (4 docs)	Law 2016-48, Circular 2019-08, Guide, Madkhal

Institutional context: albaraka.com.tn (history, 44 branches, IFN Awards); Law 2016-48 Official Journal No.58; ESPRIT Ariana (2003, CTI EUR-ACE 2023).

Appendix A - Glossary

Term	Transliteration	Meaning
al-murabaha	al-murabaha	Cost-plus sale with disclosed margin
al-ijara	al-ijara	Lease with acquisition option
al-sigha	al-sigha	Contractual form (offer & acceptance)
al-fasl	al-fasl	Article (law) / chapter (circular)
nafidh	nafidh	In force
kafin / ghayr kafin	kafin	Sufficient / Insufficient
muta'arid	muta'arid	Contradictory
RAG / hit@k / RRF	-	Retrieval-Augmented Generation / top-k / Reciprocal Rank Fusion
Citation gate	-	Quote membership + numeric gate

Appendix B - Key Figures

Fact	Source
RAGLab: 376 commits, 249 files, 86 modules	Git history & RAGLAB_FULL_REPORT.md
Corpus: 4 docs -> 339 fragments	raglab/audits/MASTER_INDEX.md
Tests: 472 unit + 133 offline, 8 workflows	CI at merge (service v1.5.0)
Web app: 255 commits, 428 files, 15 migrations	bank-assistant/docs/PROJECT_REPORT.md
Stack: NVIDIA 2048-d + Qwen (no fallback)	pipeline_policy.py

Appendix C - Reproduction

```
# RAGLab
cd RAGLab/raglab && python main.py ingest --reset --data-dir ../docs
python main.py answer "..." --query-lang ar

# Web app
cp bank-assistant/.env.docker.example .env   # set RAGLAB_URL, JWT_SECRET
docker compose up --build   # http://localhost:8080

# Verify
curl http://localhost:8000/health
curl http://localhost:8080/api/agent/health -H "Authorization: Bearer <token>"
```

Report by Farah AYARI - ESPRIT, internship at Al Baraka Bank Tunisia - Centre Urbain Nord, supervised by Mrs. Ahlem Ben Haddoud.

No identifiers or secrets are included. All figures traceable to frozen benchmark artifacts. | Internship | Centre Urbain Nord